Supplementary tables and figures

Stress-testing dark-matter substructure detectors on a common simulation suite

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These tables and figures are cited from the paper as supplementary Tables S1–S15 and Figs. S1–S3. Section, table and figure numbers inside the captions refer to the paper.

Table S1: Threshold-free comparison. Area under the ROC curve against the clean control, and completeness at 10% and 1% false-positive rate, on the common set of Table H.1, each threshold set on the 1 000-lens clean control; Table 2 gives the 1% values on the larger clean controls. The common set excludes the most massive lenses, which lowers the U-Net's AUC (0.62 here, 0.67 on all lenses). The U-Net is the mean of four seeds.

family	population	AUC	TPR at 10% FPR	TPR at 1% FPR
A joint scan	c=60	0.921	85%	78%
	c=15	0.908	81%	76%
	multipole	0.975	98%	97%
B free-form residual	c=60	0.751	53%	16%
	c=15	0.683	38%	2%
	multipole	0.897	70%	2%
C U-Net	c=60	0.620	32%	21%
	c=15	0.517	12%	3%
	multipole	0.504	10%	2%

Table S2: Family A, the joint scan, at three operating points on the full 1 000-lens Tier-0 populations. Completeness on retained fits; the paired flip counts the 831 lenses both concentration scans retain; the mass error of the localized detections agrees with that of all detections to within 0.02 dex.

	10% FPR	$\Delta\chi^2 \geq 20$	$\Delta\chi^2 \geq 100$
threshold on $\Delta\chi^2$	6.3	20.0	100.0
FPR, clean	10.0%	0.20%	0.20%
FPR, multipole a_4=0.01 θ_E	97%	86%	47%
FPR, multipole a_4=0.03 θ_E	97%	96%	90%
FPR, multipole a_3=0.03 θ_E	98%	98%	96%
compl. c=60, $10^{9.0}$–$10^{9.5} M_\odot$	98%	93%	60%
compl. c=60, $10^{9.5}$–$10^{10.0} M_\odot$	99%	95%	87%
compl. c=60, $10^{10.0}$–$10^{10.5} M_\odot$	96%	96%	94%
compl. c=60, $10^{10.5}$–$10^{11.0} M_\odot$	97%	97%	97%
compl. c=15, $10^{9.0}$–$10^{9.5} M_\odot$	95%	82%	41%
compl. c=15, $10^{9.5}$–$10^{10.0} M_\odot$	95%	92%	77%
compl. c=15, $10^{10.0}$–$10^{10.5} M_\odot$	95%	94%	89%
compl. c=15, $10^{10.5}$–$10^{11.0} M_\odot$	97%	97%	94%
paired flip c=60 → 15, lost : gained	68 : 32	47 : 9	67 : 5
detections of c=60 subhalos	710	603	475
localized (≤ 2 px)	19.9%	20.7%	21.1%
mass error, all detections (dex)	+0.39	+0.42	+0.48

Table S3: The same lenses at three depths, everything else identical. The joint scan is calibrated on the clean control of its own variant (deep: full populations; 600 s: 200-lens subsamples; shallow: 300-lens subsamples). The U-Net trained deep is scored at its deep threshold and recalibrated on each variant's control; a second set is retrained on 30 000 shallow images. [a]Four bins, 10^9–$10^{9.5}$ / $10^{9.5}$–10^{10} / 10^{10}–$10^{10.5}$ / $10^{10.5}$–10^{11} $M_\odot$, in per cent. [b]Median over the $c = 60$ population of $[\sum_{\rm pix}(I_{\rm noiseless}/\sigma)^2]^{1/2}$.

	deep	600 s	shallow
exposure time	5400 s	600 s	135 s
median arc S/N[b]	1.5×10^3	4.9×10^2	2.2×10^2
median peak-pixel S/N	235	78	37
A, joint scan			
threshold at 10% FPR	6.3	5.8	6.7
clean FPR at $\Delta\chi^2 \geq 20$	0.20%	0.00%	1.00%
macro fits rejected, c=60	70/1000	2/200	0/300
compl. c=60, 10% cal.[a]	98 / 99 / 96 / 97	76 / 96 / 95 / 93	59 / 84 / 89 / 92
compl. c=60, 10% cal., 8–8.5 / 8.5–9	47 / 79	–	8 / 9
compl. c=60, $\Delta\chi^2 \geq 20$[a]	93 / 95 / 96 / 97	52 / 89 / 93 / 93	16 / 63 / 81 / 90
compl. c=15, $\Delta\chi^2 \geq 20$[a]	82 / 92 / 94 / 97	–	6 / 37 / 58 / 88
FPR multipole, cal. / ≥ 20	97% / 96%	–	83% / 57%
C, U-Net trained deep (4 seeds)			
clean FPR at deep threshold	10%	79%	88%
AUC, c=60 vs. clean	0.67	0.58	0.52
compl. c=60, recal. 10%[a]	19 / 51 / 70 / 79	12 / 16 / 32 / 40	9 / 11 / 12 / 17
C, U-Net retrained on 30 000 shallow images (2 seeds)			
AUC, c=60 vs. clean	–	–	0.57
compl. c=60[a]	–	–	14 / 19 / 34 / 53
compl. c=15[a]	–	–	12 / 12 / 12 / 13

Table S4: Lens light added and subtracted two ways, 300 lenses per population (joint scan on the first 200), $c = 15$ template; fits with $\chi^2/\mathrm{dof} \geq 10$ excluded. "None": Tier 0 without lens light. "Single": one Sérsic fitted to the two-component truth; "double": the correctly specified fit. S1 and S2 are two independent 300-lens seed sets; the joint scan and the free-form statistic are run on S1 only. The U-Net is the four-seed mean on each set.

	none	single		double	
		S1	S2	S1	S2
smooth fit of the fixed-macro scan (shared with B)					
clean fits rejected (of 300)	5	141	148	21	12
A, joint scan (first seed set, 200 lenses; "none": full 1 000-lens populations)					
threshold, 10% FPR ($\Delta\chi^2$)	6.3	1179	–	6.2	–
FPR clean, $\Delta\chi^2 \geq 20$ / ≥ 100	0.20 / 0.20%	97 / 88%	–	0.51 / 0.51%	–
FPR multipole, $\Delta\chi^2 \geq 20$ / ≥ 100	96 / 90%	99 / 98%	–	97 / 85%	–
compl. $10^{9.5}$–$10^{10}\,M_\odot$, 10% cal.	99%	58%	–	96%	–
compl. 10^{10}–$10^{10.5}\,M_\odot$, 10% cal.	96%	100%	–	97%	–
A, fixed-macro scan (Appendix C)					
threshold, 10% FPR ($\Delta\chi^2$)	−10.6	+86	+99	−7.5	−9.3
FPR clean, $\Delta\chi^2$>20 / >100	0.68 / 0%	25 / 9%	26 / 10%	0 / 0%	0.35 / 0%
compl. 10^{10}–$10^{10.5}\,M_\odot$	85%	53%	64%	79%	89%
B, free-form residual statistic					
threshold, max $\lvert\delta\kappa\rvert$	0.10	0.96	–	0.09	–
compl. 10^{10}–$10^{10.5}\,M_\odot$	92%	12%	–	98%	–
C, U-Net (four seeds), on the subtracted images					
AUC (c=60 vs. clean)	0.67	0.58	0.57	0.67	0.66
FPR clean at Tier-0 thr.	10%	78%	80%	47%	49%
compl. 10^{10}–$10^{10.5}\,M_\odot$, recal.	70%	24%	28%	56%	60%

Table S5: $c = 60$ completeness (%) at each family's own 10% threshold on the full population, counting all detections and only those whose reported position lies within 2 pixels (0.16″) of the subhalo.

	$c = 60$ completeness (%): all detections / localized only					
family	8–8.5	8.5–9	9–9.5	9.5–10	10–10.5	10.5–11
A grid scan	47 / 11	79 / 18	98 / 20	99 / 13	96 / 17	97 / 24
A position-free scan	50 / 37	79 / 70	98 / 90	99 / 96	96 / 92	99 / 93
B free-form	15 / 0	12 / 3	45 / 31	87 / 73	96 / 66	97 / 58
C U-Net, 30 000	11 / 0	11 / 0	21 / 10	49 / 45	69 / 66	78 / 77
C U-Net, 5×10^5	9 / 2	17 / 9	38 / 32	74 / 71	84 / 84	92 / 90

Table S6: Localization on the same lenses: the 726 $c = 60$ lenses that Families A and B both retain, each family at its own 10% threshold. Per cent of detections within 2 pixels of the true subhalo, with the number of detections in parentheses, per bin of M_{200} and of the perturbation signal-to-noise. U-Net: seed 0.

		$\log_{10}(M_{200}/M_\odot)$			
family	all	8.5–9	9–9.5	9.5–10	10–10.5
A (position free)	91 (622)	89 (116)	94 (125)	98 (121)	95 (130)
B	66 (385)	29 (17)	70 (56)	84 (107)	69 (127)
C (U-Net)	76 (237)	0 (15)	50 (28)	90 (61)	94 (85)
		$\log_{10}$ S/N$_{\rm pert}$			
		1–1.5	1.5–2	2–2.5	2.5–3.5
A (position free)		90 (116)	95 (148)	96 (177)	94 (127)
B		13 (15)	67 (72)	77 (155)	66 (128)
C (U-Net)		12 (16)	55 (33)	90 (83)	92 (91)

Table S7: Line-of-sight halos, traced multi-plane, with no subhalo at the lens plane; joint scan on 300 lenses per variant, which share lenses, sources, halo masses and projected positions; only the halo redshift differs. The scan assumes $z = 0.5$. Localization and mass error are for detections at $\Delta\chi^2 \geq 20$, on the fixed grid.

line-of-sight halo redshift	foreground $z = 0.25$	background $z = 0.75$
flagged at $\Delta\chi^2 \geq 20$	58%	26%
flagged at the 10% threshold	76%	44%
macro fits rejected (of 300)	4	0
flagged at $\Delta\chi^2 \geq 20$, $10^{9.0}$–$10^{9.5} M_\odot$	52%	6%
flagged at $\Delta\chi^2 \geq 20$, $10^{9.5}$–$10^{10.0} M_\odot$	77%	12%
flagged at $\Delta\chi^2 \geq 20$, $10^{10.0}$–$10^{10.5} M_\odot$	88%	49%
flagged at $\Delta\chi^2 \geq 20$, $10^{10.5}$–$10^{11.0} M_\odot$	98%	76%
localized (≤ 2 px)	6%	0%
median offset from the halo	1.25″	0.63″
inferred minus true $\log_{10} M$ (dex)	−0.45	−1.37

Table S8: A third matched concentration. The same lenses, sources, subhalo masses and positions rendered at $c = 60$, 30 and 15, scored at each family's own 10% FPR threshold. $c = 30$ is the tidally boosted value appropriate for a subhalo near θ_E (Moliné et al. 2017). Paired flip is detections lost : gained relative to $c = 60$ on the same lenses (U-Net: mean of four seeds).

completeness at 10% FPR	$c = 60$	$c = 30$	$c = 15$
Family A, joint parametric scan			
$10^{9.0}$–$10^{9.5} M_\odot$	98%	96%	95%
$10^{9.5}$–$10^{10.0} M_\odot$	99%	96%	95%
$10^{10.0}$–$10^{10.5} M_\odot$	96%	95%	95%
$10^{10.5}$–$10^{11.0} M_\odot$	97%	96%	97%
paired flip vs. c=60	–	44 : 36	68 : 32
Family C, U-Net (mean of 4 seeds)			
$10^{9.0}$–$10^{9.5} M_\odot$	19%	10%	9%
$10^{9.5}$–$10^{10.0} M_\odot$	51%	28%	15%
$10^{10.0}$–$10^{10.5} M_\odot$	70%	44%	14%
$10^{10.5}$–$10^{11.0} M_\odot$	79%	64%	28%
paired flip vs. c=60	–	150 : 34	278 : 38

Table S9: The U-Net trained at three concentrations with one recipe and no hyperparameter search. Three training populations of 30 000 images generated from the same seed, so that only the subhalo concentration differs, each scored on the same Tier-0 test populations at its own 10% threshold. Mean and standard deviation over seeds. The two low-concentration training sets give no detector (Sect. 4.1), so their multipole row is uninformative: a network that does not detect subhalos cannot be confused by a confounder.

training set	c=60	c=15	c–M
seeds	4	2	2
AUC, c=60 vs. clean	0.668 ± 0.006	0.484, 0.485	0.474, 0.480
best validation loss	0.277	0.391	0.388
completeness at 10% FPR, tested at c=60			
$\log_{10} M_{200}$: 8.5–9.0	8 ± 2	10, 10	5, 8
$\log_{10} M_{200}$: 9.0–9.5	19 ± 2	9, 10	10, 10
$\log_{10} M_{200}$: 9.5–10.0	51 ± 5	13, 19	11, 13
$\log_{10} M_{200}$: 10.0–10.5	70 ± 1	11, 13	9, 10
$\log_{10} M_{200}$: 10.5–11.0	79 ± 1	6, 7	3, 5
completeness at 10% FPR, tested at c=15			
$\log_{10} M_{200}$: 8.5–9.0	9 ± 1	10, 11	6, 9
$\log_{10} M_{200}$: 9.0–9.5	9 ± 2	9, 10	9, 10
$\log_{10} M_{200}$: 9.5–10.0	15 ± 4	14, 19	11, 14
$\log_{10} M_{200}$: 10.0–10.5	14 ± 3	11, 13	9, 10
$\log_{10} M_{200}$: 10.5–11.0	28 ± 4	7, 7	6, 7
paired flip c=60 → 15, lost : gained	278 : 38	2 : 4	2 : 6
FPR, multipole a_4=0.03 θ_E	10 ± 1	12, 13	10, 11

Table S10: U-Net training objectives, each trained on the same 30 000-image populations at $c = 60$ and at $c = 15$ and scored by the AUC of the $c = 60$ test population against the clean control; two seeds are given as their two values, four as mean and standard deviation.

objective	trained at c=60	trained at c=15
disc, $r = 2$ px (the recipe)	0.668 ± 0.006 (4)	0.484, 0.485
disc, $r = 5$ px	0.678	0.480, 0.480
image level, max logit	0.505	0.519, 0.519
image level, log-sum-exp	0.509	0.502, 0.506

Table S11: The $c = 15$ minus $c = 60$ completeness difference at equal signal (percentage points, averaged over bins holding at least 20 lenses of each population) and the pooled two-proportion χ^2 per bin, on the full 1 000-lens populations. p is the unpaired tail probability; p_{perm} swaps each paired lens's two variants and is the one to read. U-Net: seed 0, every lens.

family	signal variable	bins	mean Δ	χ^2/dof	p	p_{perm}
A joint scan	$\log_{10} M_{\mathrm{proj}}(< 0.1'')$	3	+10	11.7	< 0.001	< 0.001
	$\log_{10} M_{\mathrm{proj}}(< 0.2'')$	4	+1	0.8	0.515	0.600
	$\log_{10}$ S/N$_{\mathrm{pert}}$	11	−6	1.8	0.048	0.005
B free-form residual	$\log_{10} M_{\mathrm{proj}}(< 0.1'')$	3	+4	4.3	0.005	0.027
	$\log_{10} M_{\mathrm{proj}}(< 0.2'')$	4	−5	2.0	0.097	0.157
	$\log_{10}$ S/N$_{\mathrm{pert}}$	10	−16	14.4	< 0.001	< 0.001
C U-Net	$\log_{10} M_{\mathrm{proj}}(< 0.1'')$	3	−7	9.0	< 0.001	< 0.001
	$\log_{10} M_{\mathrm{proj}}(< 0.2'')$	4	−12	17.8	< 0.001	< 0.001
	$\log_{10}$ S/N$_{\mathrm{pert}}$	11	−23	24.4	< 0.001	< 0.001

Table S12: The paired permutation p of the concentration-coincidence test against the TNFW mass projected within each aperture, full 1 000-lens populations. Bold: rejected at family-wise $\alpha = 0.05$ after Holm's correction for the seven apertures. U-Net: seed 0.

aperture	0.05″	0.075″	0.1″	0.15″	0.2″	0.3″	0.4″
A, joint scan	**<0.001**	**<0.001**	**<0.001**	0.050	0.600	0.695	0.022
B, free-form	**<0.001**	**<0.001**	0.027	0.411	0.157	**<0.001**	**<0.001**
C, U-Net	0.404	**<0.001**	**<0.001**	**<0.001**	**<0.001**	**<0.001**	**<0.001**

Table S13: The fit-quality cut of Sect. 2.3 on the full 1 000-lens joint-scan populations: lenses whose first smooth start fails $\chi^2/\mathrm{dof} < 10$; of those, the number retained because a smooth re-fit (from a joint solution with the perturber removed, or from the seven-start re-fit) reaches the fit-quality cut, and because only the joint model does; and the number excluded, with the excluded subhalo-bearing lenses below $10^{10}\,M_\odot$.

population	first start	smooth re-fit	joint only	excluded	excl. $< 10^{10}\,M_\odot$
clean control	37	37	0	0	–
$c = 60$	167	33	64	70	2
$c = 30$	110	33	48	29	2
$c = 15$	74	38	23	13	0
$m = 4$, $a_4/\theta_\mathrm{E} = 0.01$	56	37	0	19	–
$m = 4$, $a_4/\theta_\mathrm{E} = 0.03$	106	40	19	47	–
$m = 3$, $a_3/\theta_\mathrm{E} = 0.03$	201	29	89	83	–

Table S14: The perturber-template control for the joint scan: the loss in $c = 60$ minus $c = 15$ completeness, in points, on the same 300-lens subsample, for the $c = 15$ template used throughout, the $c = 60$ template and the Dutton & Macciò (2014) relation, each calibrated on the same 300-lens clean control (thr. is its 10% threshold on $\Delta\chi^2$).

template	thr.	8–8.5	8.5–9	9–9.5	9.5–10	10–10.5
loss c=60 → 15 *at the 10% threshold (points)*						
$c = 15$ (as run)	6.6	+21	+4	+3	+5	+3
$c = 60$	6.5	+11	+1	−3	+2	−0
c–M	6.5	+23	+7	−3	+2	−0
loss at $\Delta\chi^2 \geq 20$ *(points)*						
$c = 15$ (as run)		+4	+7	+9	+2	+5
$c = 60$		+8	+13	+6	−0	+2
c–M		+4	+10	+3	−0	+2

Table S15: Concentration and truncation varied together, fixed-macro scan, rates at the null-floored $\Delta\chi^2 > 0$ (Appendix C). The upper block holds $\tau = 20$ fixed while varying c, as elsewhere in this paper; the lower block fixes the truncation radius in physical terms so τ scales with c. Same lenses, sources, masses and positions throughout.

completeness at $\Delta\chi^2 > 0$	c=60	c=30	c=15
$\tau = 20$ fixed (this paper elsewhere, and the literature)			
$10^{9.0}$–$10^{9.5} M_\odot$	28%	23%	16%
$10^{9.5}$–$10^{10.0} M_\odot$	69%	55%	47%
$10^{10.0}$–$10^{10.5} M_\odot$	81%	77%	73%
τ scaled with c (tidal track: $\tau = 20, 10, 5$)			
$10^{9.0}$–$10^{9.5} M_\odot$	28%	19%	9%
$10^{9.5}$–$10^{10.0} M_\odot$	69%	54%	44%
$10^{10.0}$–$10^{10.5} M_\odot$	81%	77%	66%
total mass of the truncated profile, as a fraction of M_{200}			
$\tau = 20$ fixed	0.68	0.86	1.16
tidal track	0.68	0.63	0.57

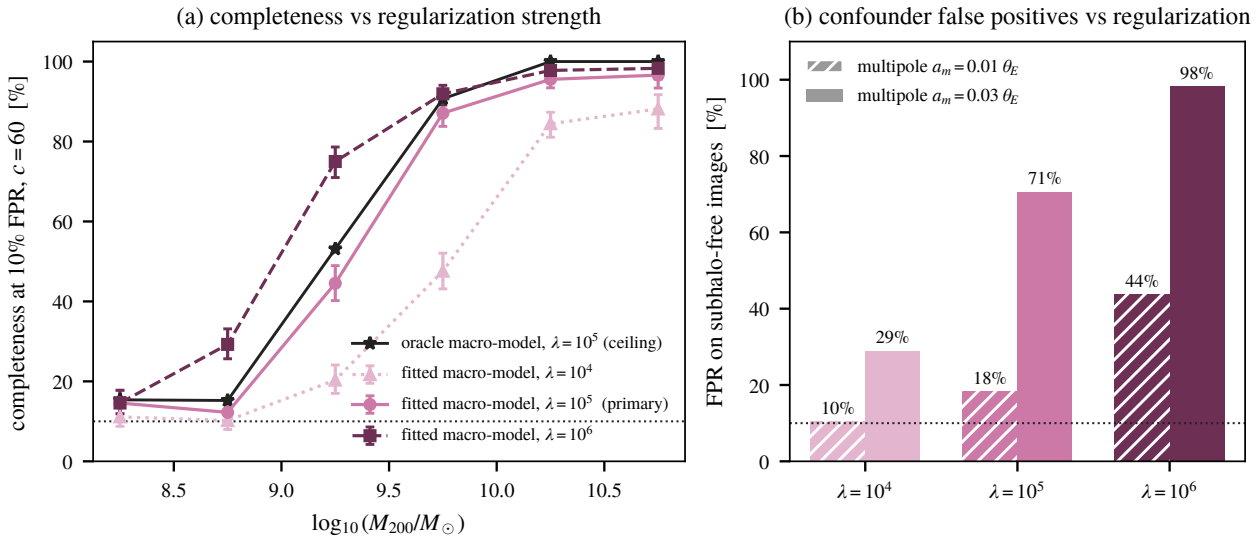


Figure S1: The free-form residual statistic's one free choice, the regularization strength λ. (a) Completeness at $c = 60$ on the full 1 000-lens populations for the fitted macro-model at three λ (10^5 is the value used throughout; 68% Wilson intervals) and, on the 300-lens seed-0 subsample, for the exact macro-model at $\lambda = 10^5$ (black stars, the ceiling at that λ; a fitted curve at another λ can exceed it). $\lambda = 10^4$, 10^5 and 10^6 in light to dark shades of one hue. At $\lambda = 10^4$ the statistic is dominated by fitted noise and its low-mass points scatter about the 10% chance line; a point below the line is a fluctuation. (b) False-positive rate on the two multipole populations at the same three λ (hatched: $a_m/\theta_{\mathrm{E}} = 0.01$).

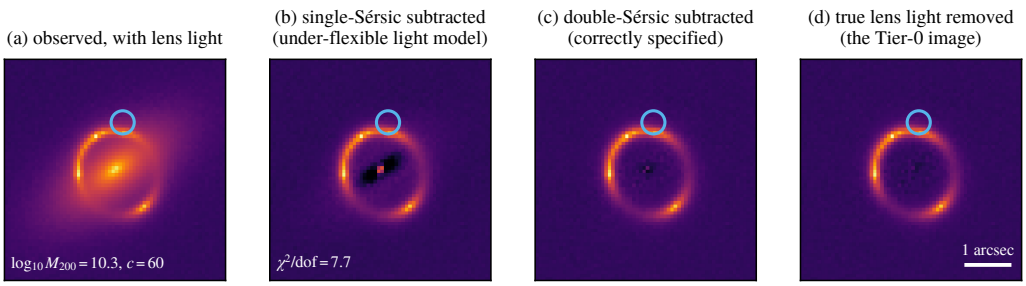


Figure S2: What lens light does to the image a detector sees. One $c = 60$ subhalo lens: (a) as observed; (b) after subtracting the single Sérsic fitted jointly with the mass model, whose misfit leaves a central residual brighter than any subhalo signature; (c) after the correctly specified double Sérsic; (d) with the true lens light removed. The circle marks the true subhalo. Same arcsinh stretch throughout.

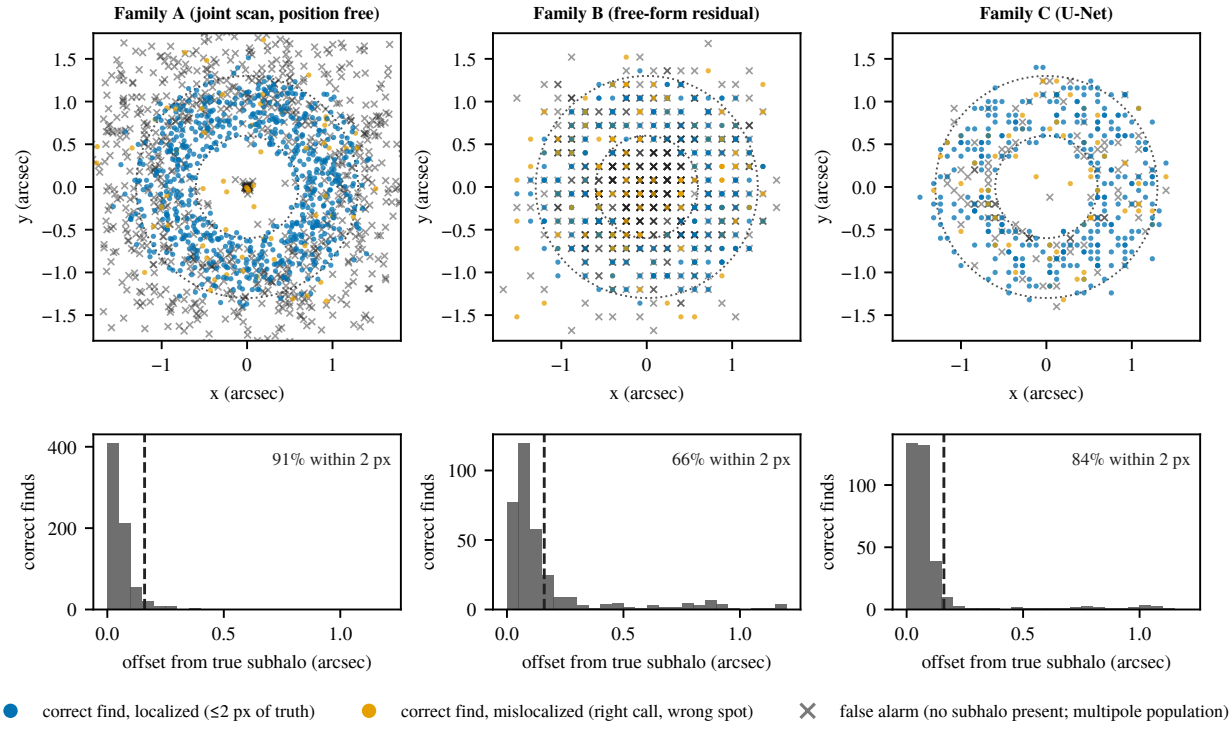


Figure S3: Where each method points ($c = 60$ population; the joint scan with the perturber position free, the free-form statistic and the U-Net seed 0), for the detections at each family's own 10% threshold. Top: the reported position of every detection, blue if within 2 pixels (0.16″) of the true subhalo, orange if not; grey ×: false alarms on the multipole population; dotted rings: the placement annulus. Bottom: offsets of the correct finds, with the 2-pixel criterion dashed. Fractions are for each family's full detection sample; supplementary Table S6 compares them on the common set.

References

Dutton, A. A. & Macciò, A. V. 2014, MNRAS, 441, 3359

Moliné, Á., Sánchez-Conde, M. A., Palomares-Ruiz, S., & Prada, F. 2017, MNRAS, 466, 4974